Titanic

April 14, 2025

[1]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

[10]:
```python
train_df = pd.read_csv('train.csv')
test_df = pd.read_csv('test.csv')
gender_submission_df = pd.read_csv('gender_submission.csv')
```

[11]:
```python
print(train_df)
```

```
     PassengerId  Survived  Pclass  \
0              1         0       3
1              2         1       1
2              3         1       3
3              4         1       1
4              5         0       3
..           ...       ...     ...
886          887         0       2
887          888         1       1
888          889         0       3
889          890         1       1
890          891         0       3

                                                  Name     Sex   Age  SibSp  \
0                              Braund, Mr. Owen Harris    male  22.0      1
1    Cumings, Mrs. John Bradley (Florence Briggs Th…  female  38.0      1
2                               Heikkinen, Miss. Laina  female  26.0      0
3         Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                             Allen, Mr. William Henry    male  35.0      0
..                                                 ...     ...   ...    ...
886                              Montvila, Rev. Juozas    male  27.0      0
887                       Graham, Miss. Margaret Edith  female  19.0      0
888           Johnston, Miss. Catherine Helen "Carrie"  female   NaN      1
889                              Behr, Mr. Karl Howell    male  26.0      0
890                                Dooley, Mr. Patrick    male  32.0      0

     Parch            Ticket     Fare Cabin Embarked
```

```
0         0          A/5 21171   7.2500   NaN        S
1         0           PC 17599  71.2833   C85        C
2         0   STON/O2. 3101282   7.9250   NaN        S
3         0             113803  53.1000  C123        S
4         0             373450   8.0500   NaN        S
..      …                  …        …    …      …
886       0             211536  13.0000   NaN        S
887       0             112053  30.0000   B42        S
888       2         W./C. 6607  23.4500   NaN        S
889       0             111369  30.0000  C148        C
890       0             370376   7.7500   NaN        Q

[891 rows x 12 columns]
```

[20]:
```
test_df.head()
```

[20]:
```
   PassengerId  Pclass                                          Name     Sex  \
0          892       3                              Kelly, Mr. James    male
1          893       3              Wilkes, Mrs. James (Ellen Needs)  female
2          894       2                     Myles, Mr. Thomas Francis    male
3          895       3                              Wirz, Mr. Albert    male
4          896       3  Hirvonen, Mrs. Alexander (Helga E Lindqvist)  female

    Age  SibSp  Parch   Ticket     Fare Cabin Embarked
0  34.5      0      0   330911   7.8292   NaN        Q
1  47.0      1      0   363272   7.0000   NaN        S
2  62.0      0      0   240276   9.6875   NaN        Q
3  27.0      0      0   315154   8.6625   NaN        S
4  22.0      1      1  3101298  12.2875   NaN        S
```

[21]:
```
gender_submission_df.head()
```

[21]:
```
   PassengerId  Survived
0          892         0
1          893         1
2          894         0
3          895         0
4          896         1
```

[28]:
```
train_df.shape
```

[28]:
```
(891, 12)
```

[22]:
```
train_df.head()
```

[22]:
```
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
```

```
2            3          1       3
3            4          1       1
4            5          0       3

                                               Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th…  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
```

[18]:
```
train_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

[19]:
```
train_df.describe()
```

[19]:
```
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
```

```
50%      446.000000    0.000000    3.000000   28.000000    0.000000
75%      668.500000    1.000000    3.000000   38.000000    1.000000
max      891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
```

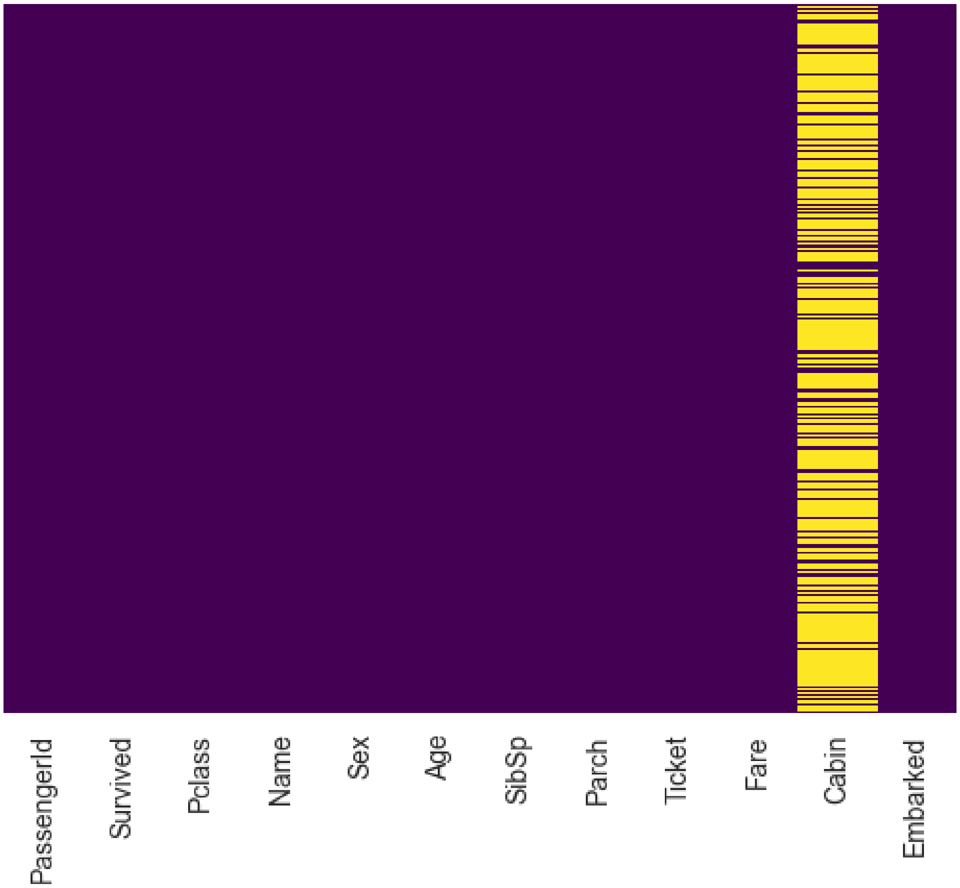
Exploratory Data Analysis

[24]:
```
train_df.isnull().sum()
```

[24]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

[60]:
```
sns.heatmap(train_df.isnull(),yticklabels=False,cbar=False,cmap="viridis")
```

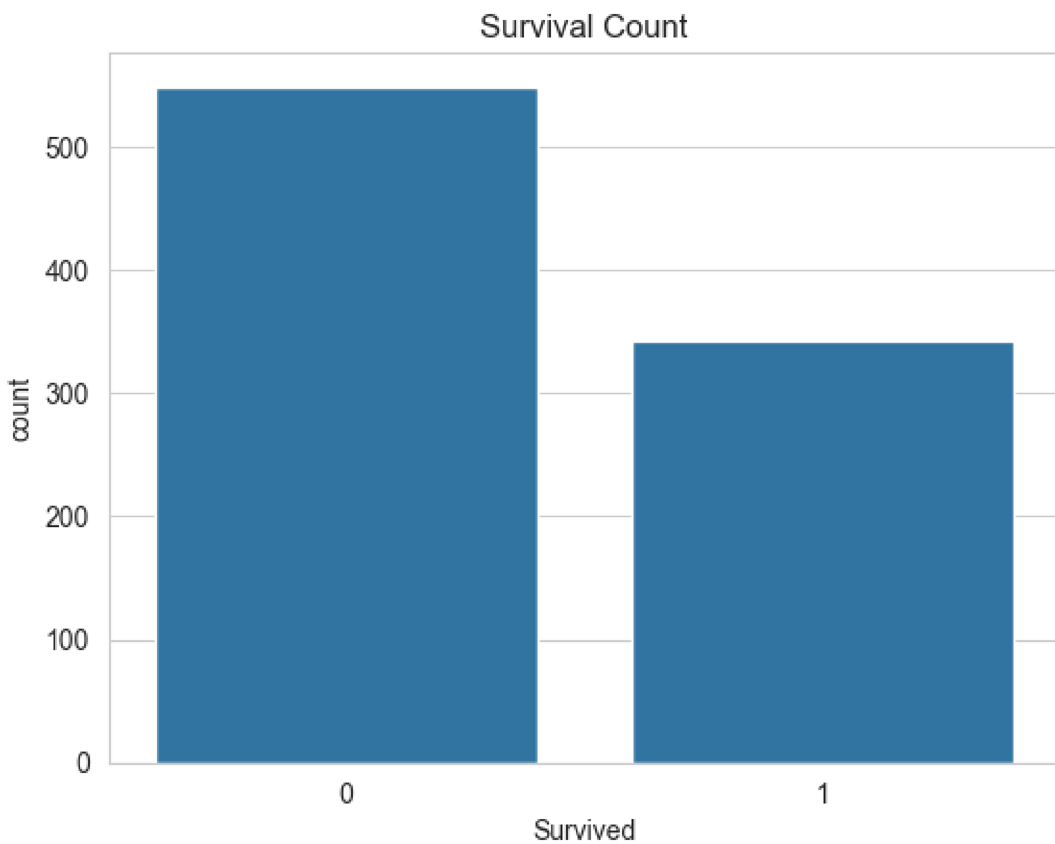
[60]:
```
<Axes: >
```



Roughly 20 percent of the Age data is missing.The proportion of Age missing is likely small enough for reasonable replacement with some form of imputation. Looking at the Cabin column,it looks like we are just missing too much of that data to do something useful with at a basic level.We'll probably drop this later,or change it to another feature like 'Cabin Known:0 or

[29]:
```
sns.set_style('whitegrid')
sns.countplot(x='Survived', data=train_df)
plt.title('Survival Count')
```

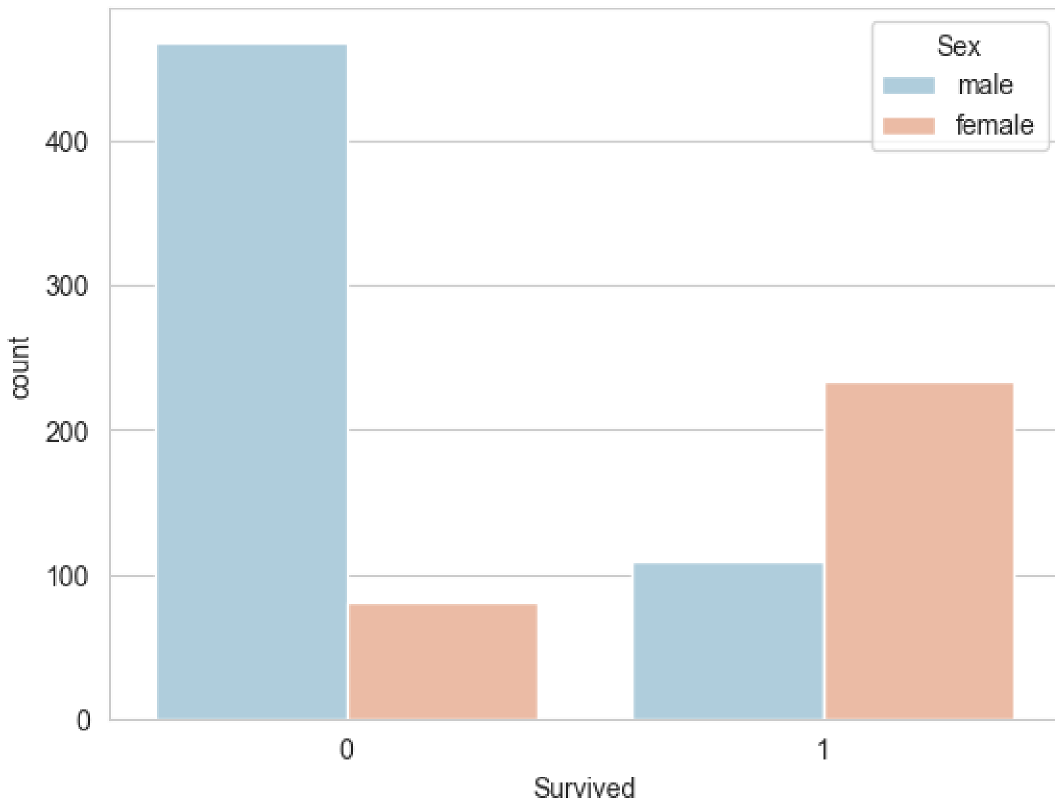
[29]:
```
Text(0.5, 1.0, 'Survival Count')
```



Observation: There were more passengers who did not survive (0) more than 550 than those who did (1). that is more than 300.

[31]:
```
sns.set_style('whitegrid')
sns.countplot(x='Survived', hue='Sex', data=train_df, palette="RdBu_r")
```

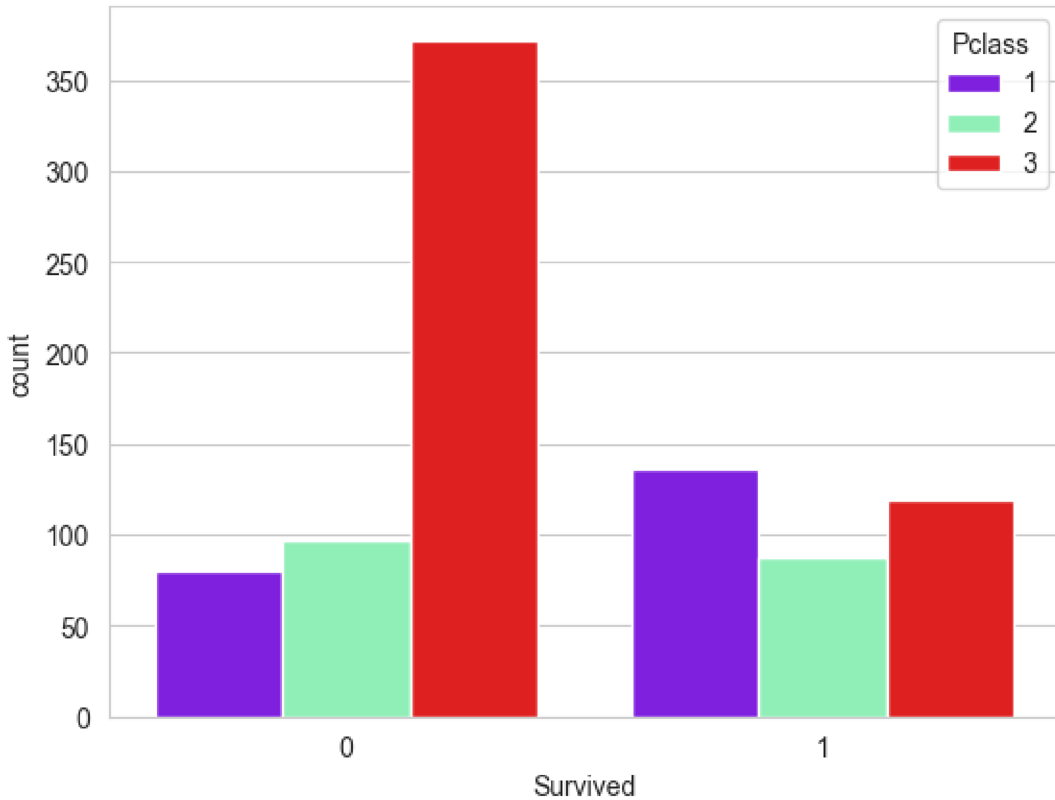
[31]: <Axes: xlabel='Survived', ylabel='count'>



Observation: most of the people those are male that is more than 450 is not servived.Wheareas significantly higher number of females survived compared to males. This suggests that women were given priority during evacuation.

[32]:
```
sns.set_style('whitegrid')
sns.countplot(x='Survived', hue='Pclass',data=train_df,palette='rainbow')
```

[32]:
```
<Axes: xlabel='Survived', ylabel='count'>
```

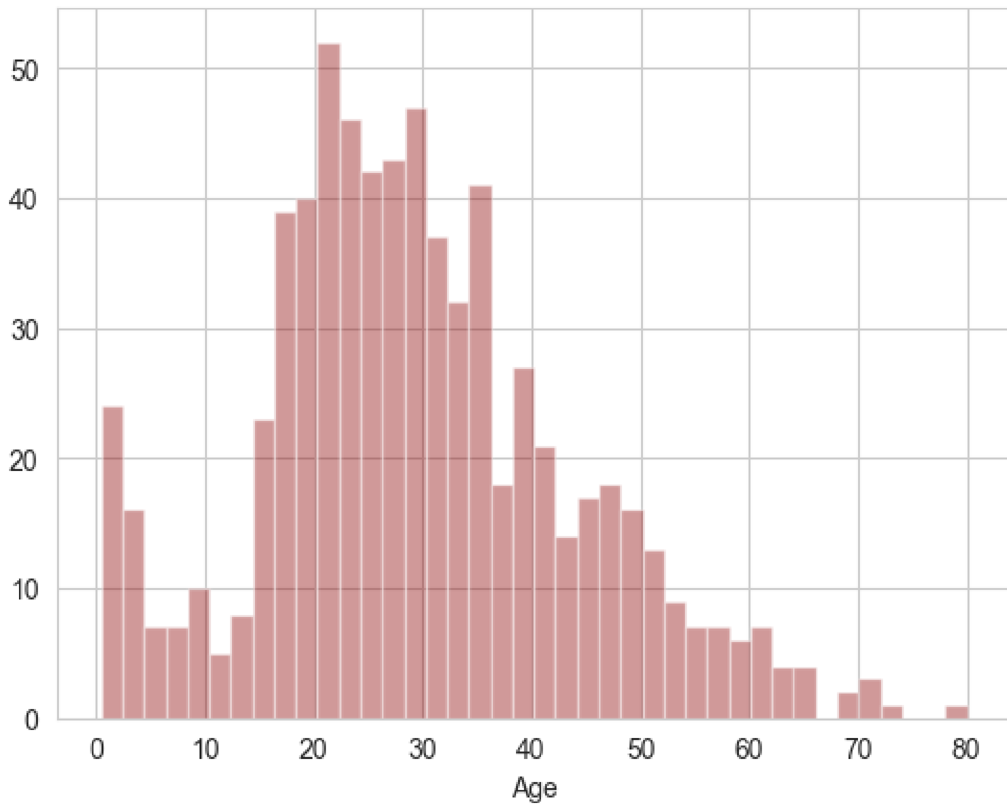


Observation: Passengers in 1st class had a much higher survival rate compared to those in 2nd and especially 3rd class. This shows that class status played a role in survival chances.

[36]:
```
sns.distplot(train_df["Age"].dropna(),kde=False,color="darkred",bins=40)
```

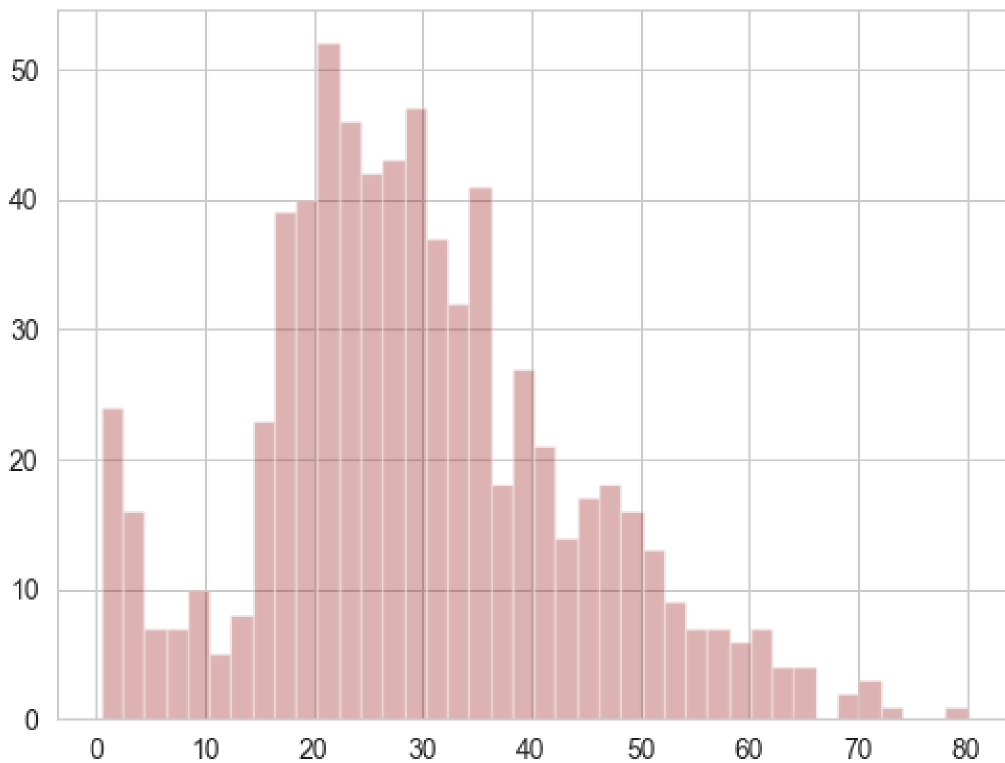
```
C:\Users\Pooja\AppData\Local\Temp\ipykernel_14016\4102928446.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(train_df["Age"].dropna(),kde=False,color="darkred",bins=40)
```

[36]: <Axes: xlabel='Age'>




```
[37]: train_df["Age"].hist(color="darkred",bins=40,alpha=0.3)
```

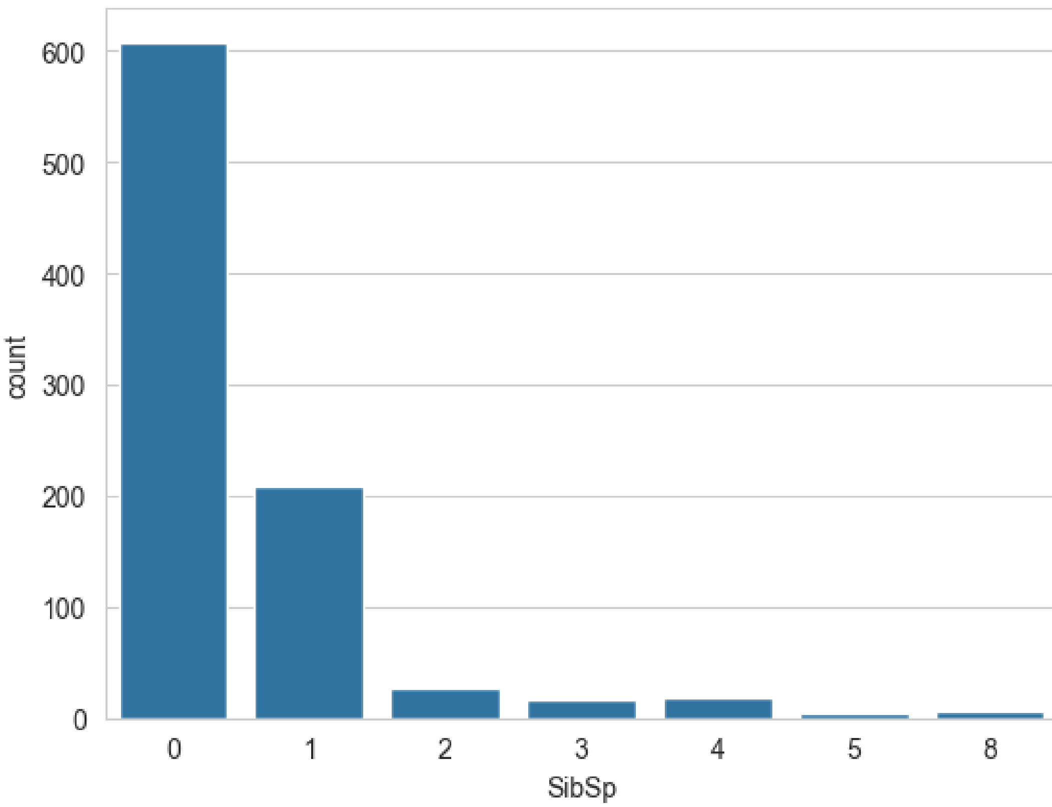
```
[37]: <Axes: >
```



Observation: Most passengers were between 20 and 40 years old. Very few passengers were above 60, and there were some children as well.

[38]:
```
sns.countplot(x='SibSp',data=train_df)
```

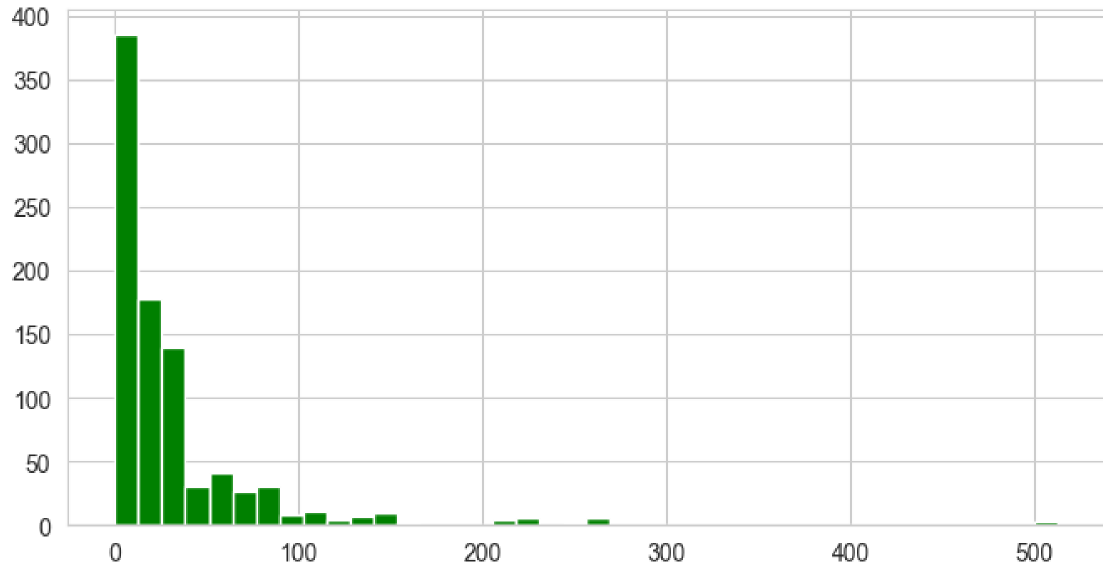
[38]:
```
<Axes: xlabel='SibSp', ylabel='count'>
```



Observation: The majority of passengers were traveling alone or with one sibling/spouse. Very few had more than 2 siblings/spouses with them.

[64]:
```
train_df['Fare'].hist(color='green',bins=40,figsize=(8,4))
```

[64]:
```
<Axes: >
```



Observation: Most passengers paid a fare under \$100. A few paid significantly higher fares, which likely corresponds to first-class tickets.

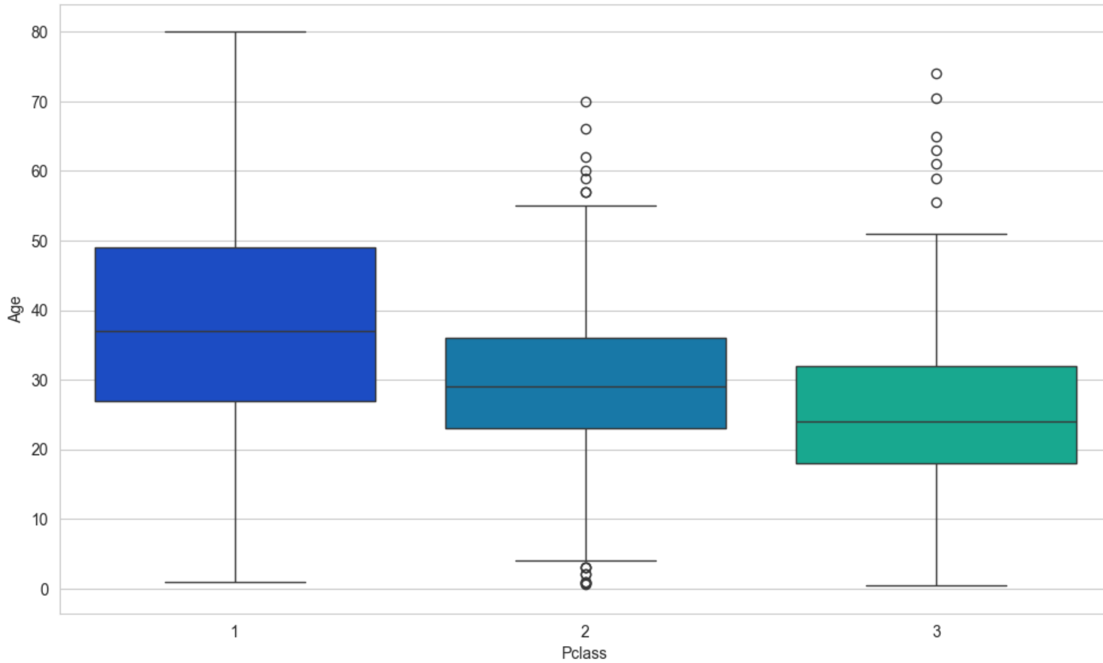
Data Cleaning

We want to fill in missing Age data instead od just dropping the missing Age data rows.One way to do this by filling in the mean age of all the passangers(imputation).

[41]:
```
plt.figure(figsize=(12,7))
sns.boxplot(x='Pclass',y='Age',data=train_df,palette='winter')
```

```
C:\Users\Pooja\AppData\Local\Temp\ipykernel_14016\3472948166.py:2:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.boxplot(x='Pclass',y='Age',data=train_df,palette='winter')
```

[41]:
```
<Axes: xlabel='Pclass', ylabel='Age'>
```



we can see that the wealthier passagers in the higher classes trend to be older,which makes sense. we'll use these average age values to impute based on Pclass for Age.

[55]:
```
def impute_age(cols):
    Age = cols[0]
    Pclass = cols[1]

    if pd.isnull(Age):

        if Pclass == 1:
            return 37

        elif Pclass == 2:
            return 29

        else:
            return 24
    else:
        return Age
```

Now apply that funtional

[56]:
```
train_df['Age'] = train_df[['Age','Pclass']].apply(impute_age,axis=1)
```

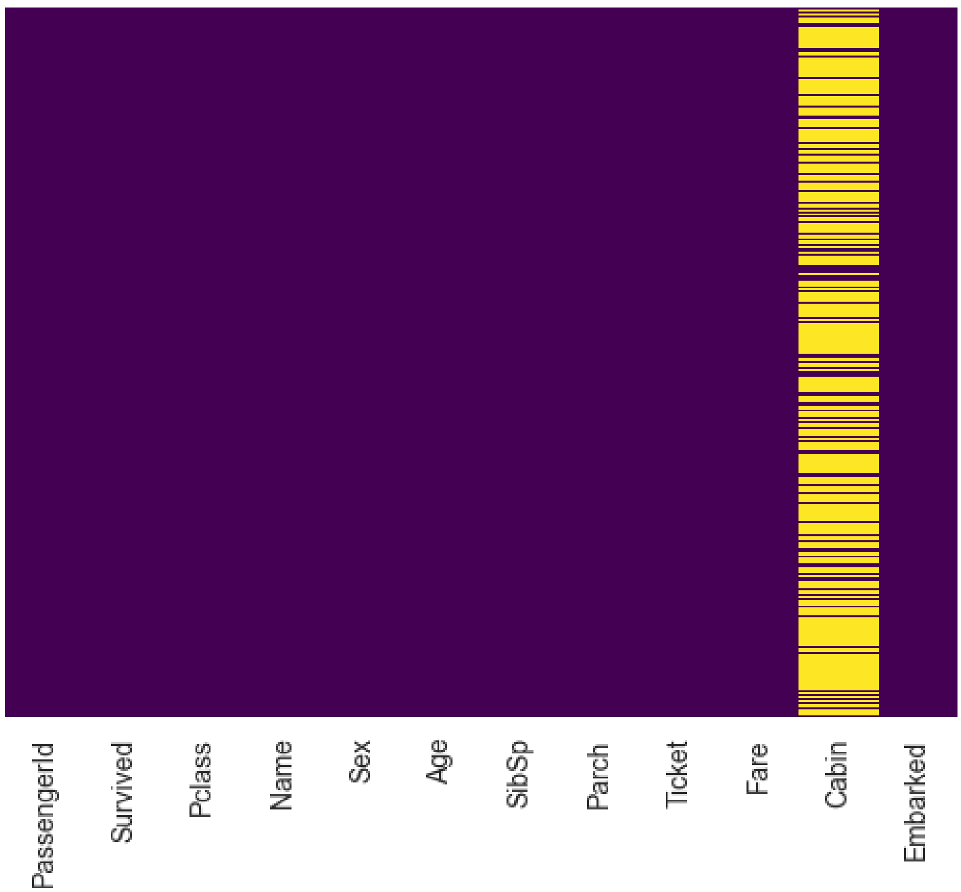
```
C:\Users\Pooja\AppData\Local\Temp\ipykernel_14016\3005635740.py:2:
FutureWarning: Series.__getitem__ treating keys as positions is deprecated. In a
```

```
future version, integer keys will always be treated as labels (consistent with
DataFrame behavior). To access a value by position, use `ser.iloc[pos]`
  Age = cols[0]
C:\Users\Pooja\AppData\Local\Temp\ipykernel_14016\3005635740.py:3:
FutureWarning: Series.__getitem__ treating keys as positions is deprecated. In a
future version, integer keys will always be treated as labels (consistent with
DataFrame behavior). To access a value by position, use `ser.iloc[pos]`
  Pclass = cols[1]
```

Now check that heat map again

[57]:
```
sns.heatmap(train_df.isnull(),yticklabels=False,cbar=False,cmap="viridis")
```

[57]: <Axes: >



[61]:
```
train_df.drop('Cabin',axis=1,inplace=True)
```

[62]:
```
train_df.head()
```

[62]:

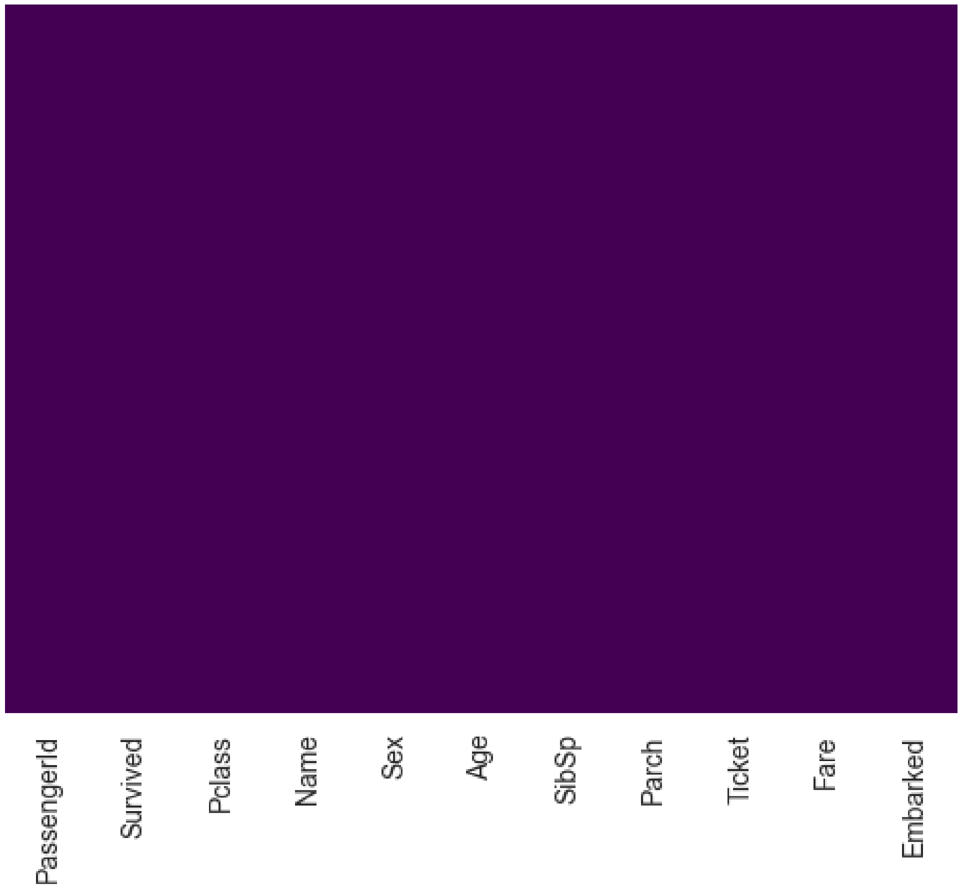
```
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  24.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th…  female  37.0      1
2                             Heikkinen, Miss. Laina  female  24.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  37.0      1
4                           Allen, Mr. William Henry    male  24.0      0

   Parch            Ticket     Fare Embarked
0      0         A/5 21171   7.2500        S
1      0          PC 17599  71.2833        C
2      0  STON/O2. 3101282   7.9250        S
3      0            113803  53.1000        S
4      0            373450   8.0500        S
```

[63]:

```
sns.heatmap(train_df.isnull(),yticklabels=False,cbar=False,cmap="viridis")
```

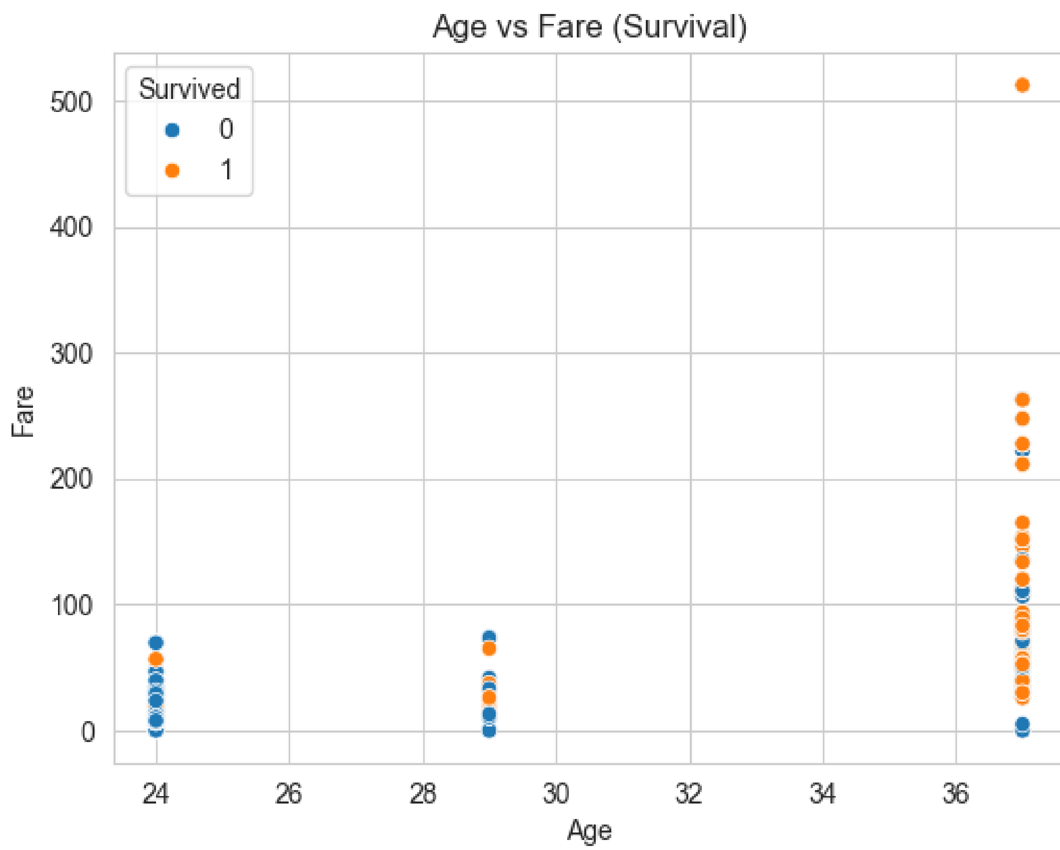
[63]:

```
<Axes: >
```



[65]:
```
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=train_df)
plt.title('Age vs Fare (Survival)')
```

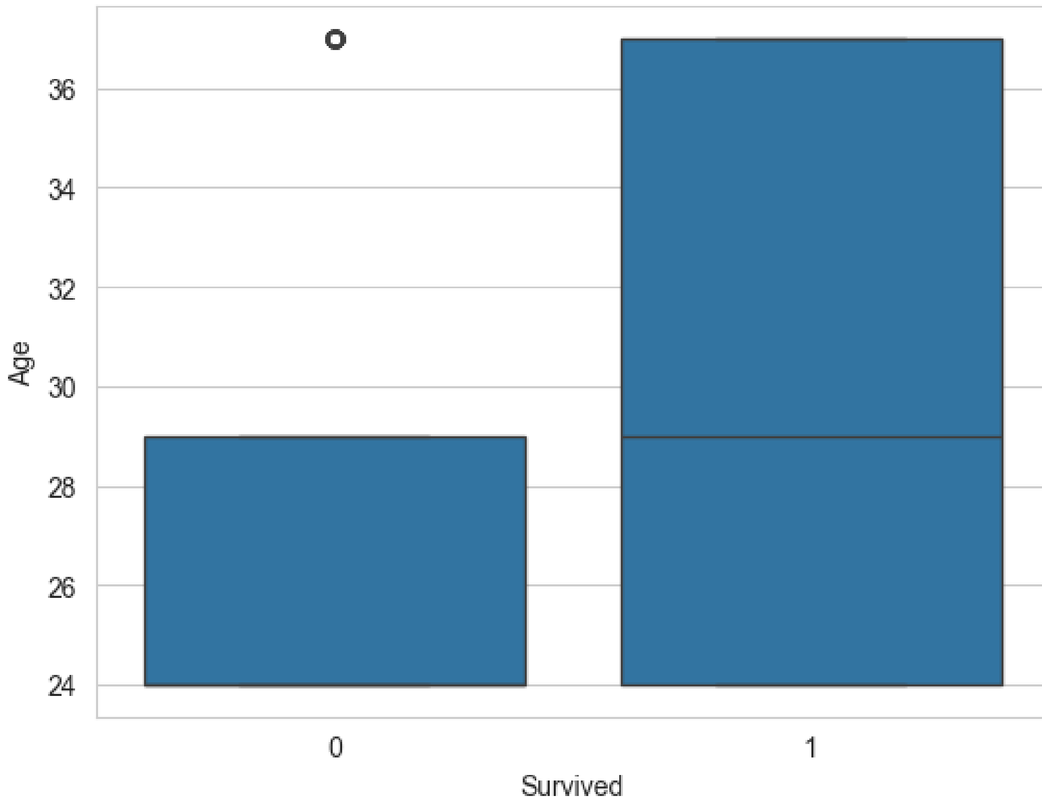
[65]: Text(0.5, 1.0, 'Age vs Fare (Survival)')



Observation: Passengers who paid higher fares were more likely to survive, especially among younger age groups. Survivors are clustered more in higher fare zones, while non-survivors are spread across all fare ranges.

[68]:
```
sns.boxplot(x='Survived', y='Age', data=train_df)
```

[68]: <Axes: xlabel='Survived', ylabel='Age'>



Observation: The median age of survivors is slightly lower than that of non-survivors. The interquartile range for survivors is also narrower, suggesting survivors were more likely to be younger, but people of all ages were affected.

###Conclusion : Survival on the Titanic was influenced by a combination of gender, passenger class, and fare, with females and wealthier (1st-class) passengers having the highest survival odds. Age had a moderate effect, with younger passengers showing slightly better survival rates.

[ ]:

```

```